

MARIOS PAPAMICHALIS

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RESEARCH INTERESTS

Statistical machine learning and the evaluation of AI systems. Bayesian statistics. Statistical network analysis. Robustness and sensitivity analysis. Causal inference. Applied statistics in health, finance and the social sciences.

ACADEMIC APPOINTMENTS

Yale University, Human Nature Lab and Yale Institute for Network Science (*Sep. 2021 to present*)

Postdoctoral Associate, Research Fellow and Affiliated Researcher. Supervisor: Prof. Nicholas A. Christakis

Large-scale sociocentric network studies and randomized interventions in rural Honduras, causal inference for network interventions, statistical modeling of multiplex and signed networks.

Temple University (*Sep. 2020 to Aug. 2021*)

Postdoctoral Researcher. Supervisor: Prof. Edoardo M. Airolidi. Visiting Scholar, Yale University.

Purdue University (*Feb. 2019 to Aug. 2020*)

Postdoctoral Researcher. Supervisor: Prof. Patrick J. Wolfe

EDUCATION

University College London (UCL) (*Feb. 2015 to Feb. 2019*)

MPhil and PhD in Statistics (Distinction). Thesis: Sampling Designs and Robustness on Networks. Visiting Scholar, Purdue University.

University College London (UCL) (*Sep. 2011 to Sep. 2012*)

MSc in Machine Learning and Computational Statistics (Distinction). Dissertation: Time Series Models Driven by Fractional Brownian Motion.

National Technical University of Athens (*Sep. 2005 to Jun. 2010*)

MEng and BEng in Electrical and Computer Engineering (Distinction). Diploma thesis: Evaluation of Probabilistic Region Queries for Uncertain Positions of Moving Objects.

PUBLICATIONS: ACCEPTED, IN PRESS AND PUBLISHED

“Educational Intervention Can Rewire Village Social Networks” (2026)

M. Papamichalis, L. Forastiere*, E. M. Airolidi and N. A. Christakis. *Joint first authors.*

Accepted, *Nature*.

Contribution. Led the statistical analyses, and wrote and archived the analysis code.

Abstract. Using a two-stage randomized public-health intervention in 110 remote Honduran villages (8,331 individuals), we quantify how an informational intervention alters pre-existing sociocentric networks. In low-dosage villages, treated individuals tend to sever health-advice ties with untreated alters, while in high-dosage villages treated individuals increase both inbound and outbound ties. The intervention also affects friendship and financial ties, indicating multiplex spillovers, yet aggregate village-level structure changes only modestly.

“Support-Safe Variational Hybrid Filtering for Contact-Mode and Sparse-Law Recovery” (2026)

M. Papamichalis and R. Ruane.

Accepted, *Advances in Neural Information Processing Systems (NeurIPS)*. [arXiv:2605.16398](https://arxiv.org/abs/2605.16398).

Abstract. Variational filtering for hybrid dynamical systems in which the active discrete mode and a sparse governing law must be recovered jointly. Every iterate is constrained to the admissible support, so non-negativity, complementarity and sparsity hold by construction rather than by penalty.

“Social Network Structure and Depression by Gender in Rural Honduras: A Cross-Sectional Study” (2026)
S. T. Lee, M. Papamichalis*, K. Raygoza-Cortez, N. A. Christakis and A. L. Rodríguez de la Rosa.* *Joint first authors.

BMJ Open, 16(6), e108285. Open access.

Contribution. Methodological lead. Designed and built the statistical analysis.

“Probabilistic Range Monitoring of Streaming Uncertain Positions in Geosocial Networks” (2012)

K. Patroumpas, M. Papamichalis and T. Sellis.

Proceedings of SSDBM 2012, 24th International Conference on Scientific and Statistical Database Management. **Best Paper Award.**

MANUSCRIPTS UNDER REVISION

“Bayesian Predictive Synthesis for Dynamic Networks: Forecasting and Identifying Structural Mechanisms” (2026)

M. Papamichalis, R. Ruane and T. Papamichalis.

Under revision, 2nd R&R, *Journal of the American Statistical Association*. [arXiv:2606.26136](https://arxiv.org/abs/2606.26136).

Abstract. We combine competing generative models for dynamic networks through a predictive synthesis framework defined at the level of the graphon, so that forecasts are calibrated and the weights assigned to component models are interpretable as evidence for competing structural mechanisms of tie formation and dissolution.

“Clustering Network Formation in Populations of Networks” (2026)

M. Papamichalis, E. M. Airolidi, N. A. Christakis and L. Forastiere.

Under revision, 1st R&R, *Annals of Applied Statistics*.

Abstract. Inference for populations of networks with different node sets and sizes, where subsets of networks share distinct formation regimes. Each observed network is modeled as a noisy snapshot of an underlying one, and networks are clustered jointly by formation mechanism and by measurement-error profile, with parallelizable block-updated MCMC and exchange-type updates.

“Decision-Theoretic Robustness for Network Models” (2026)

M. Papamichalis, R. Ruane, S. Chandna and S. Lunagómez.

Under revision, *Journal of Machine Learning Research*.

Abstract. A decision-theoretic notion of robustness obtained by letting the posterior vary within a small Kullback–Leibler neighborhood. For exchangeable graphs we connect robust posterior risk to graphon limits, which yields tractable sensitivity expansions computed by exponential tilting of posterior draws already generated.

“Deep Description of Static and Dynamic Network Ties in Honduran Villages” (2026)

M. Papamichalis, N. Nakis and N. A. Christakis.

Under revision, R&R, *Journal of Complex Networks*.

MANUSCRIPTS UNDER REVIEW

“Network Changes in Rural Honduras: Educational Intervention via Strategic Targeting” (2026)

M. Papamichalis, L. Forastiere, E. M. Airolidi and N. A. Christakis. *Journal of Complex Networks*.

“Spatial Structure of Friendly and Antagonistic Ties in Honduran Villages” (2026)

M. Papamichalis, M. L. Traeger and N. A. Christakis. *Journal of Complex Networks*.

“Social Factors Associated with Generalized Anxiety Disorder in Honduran Villages” (2026)

M. Papamichalis, A. L. Rodríguez de la Rosa and N. A. Christakis. *BMJ Open*.

WORKING PAPERS

“Latent Space Network Modelling with Hyperbolic and Spherical Geometries” (2026)

K. Turnbull, M. Papamichalis, S. Lunagómez and E. M. Airolidi.

Identifiability of latent representations under non-Euclidean curvature, with MCMC and variational estimation.

“Social and Financial Ties Are Directed Toward Larger People in Rural Honduran Villages” (2026)

M. Papamichalis, A. L. Rodríguez de la Rosa and N. A. Christakis.

Six complete directed networks across 2,444 adults in 19 villages, tested against a within-village permutation null.

“Peer Effects in Repeated Cooperation Games in Isolated Honduran Villages” (2026)

M. Papamichalis, T. Papamichalis, N. A. Christakis and F. Fu.

Lab-in-the-field public-goods experiment across 134 villages with 2,591 participants and a complete pre-play friendship network.

“Intermediary Black–Scholes: Who Insures the Crash, and What the Crash Costs” (2026)

M. Papamichalis and T. Papamichalis.

Crash insurance pricing at a leverage threshold in a production economy with rare disasters.

“Forgetting Friends and Foes: Self-Reported Errors in Sociocentrically Mapped Village Networks”

M. Papamichalis, A. L. Rodríguez de la Rosa, K. Raygoza-Cortez and N. A. Christakis.

Systematic reporting error in a large sociocentric census, and which conclusions survive it.

“Attention Concentration in Open-Weight Language Models”

M. Papamichalis.

Attention maps treated as weighted directed graphs. Concentration of attention mass, its stability across layers and prompts, and the effect on quantities estimated downstream.

“DeFi Statistical Network Analysis (Ethereum and On-Chain Networks)”

M. Papamichalis, L. Tassioulas and collaborators, Yale University.

Network evolution, structural breaks and fragility across decentralized finance protocols and market regimes.

AWARDS AND HONORS

Best Paper Award, SSDBM 2012 (2012)

24th International Conference on Scientific and Statistical Database Management, Chania, Crete, Greece.

Gold Medal, Balkan Mathematical Olympiad (2005)

Bronze Medal, International Mathematical Olympiad (IMO) (2004)

Bronze Medal, Balkan Mathematical Olympiad (2004)

TEACHING EXPERIENCE

Guest Lecturer, STAT 4750/9200, Sample Survey Methods (Nov. 2025)

The Wharton School, University of Pennsylvania. Lecture and paper presentation on the design and analysis of a randomized village-network study.

Teaching Assistant, Causal Inference (graduate) (2020 to 2021)

Temple University, with Prof. Edoardo M. Airolidi.

Teaching Assistant and Course Support, Machine Learning I (2019 to 2020)

Purdue University.

Teaching Assistant, Further Probability and Statistics (Oct. to Dec. 2017)

University College London.

Teaching Assistant, Introduction to Statistical Methods and Computing (MPhil) (Oct. to Dec. 2016)

University College London.

Additional UCL teaching assistantships (2015 to 2018)

Probability and Statistics II (2nd year), Optimization Algorithms in Operational Research (2nd year), Statistical Inference (3rd year), Forecasting (3rd year).

Assistant Teacher, Mathematics (high school) (Oct. 2006 to May 2007)

Athens College High School, Filothei, Athens, Greece.

COURSES PREPARED TO TEACH

Core and introductory: calculus, linear algebra, introductory probability and statistics, statistical computing.

Statistics: mathematical statistics, statistical inference, regression and linear models, design of experiments, time series and forecasting, Bayesian statistics.

Machine learning and AI: machine learning, statistical learning, data science, evaluation and reliability of AI systems, AI and society.

Electives and projects: causal inference and program evaluation, statistical methods for network data, senior thesis, capstone and Clinic supervision.

RESEARCH SUPERVISION AND MENTORING

Research mentoring of medical students, doctoral researchers and postdoctoral colleagues at the Human Nature Lab, Yale University, on study design, statistical and causal methodology, reproducible computation and scientific writing, with co-authorship in each case. Mentoring of junior analysts at Wells Fargo on model development, evaluation design and communication of results to non-technical decision makers.

APPLIED AND INDUSTRY EXPERIENCE

Eleven years of applied statistical and machine learning practice alongside academic research, spanning technology, quantitative finance and energy.

Wells Fargo, Quantitative Strategy (Corporate and Investment Banking) (*Dec. 2020 to Jun. 2023*)

Quantitative Analyst (part-time), Manhattan, NY.

Volume-prediction trading strategies using dynamic linear models, hierarchical models for market distress detection, order book distribution analysis, and indicators for dark pool activity.

Meta (Facebook) (*May to Aug. 2018*)

Statistician (contract), San Francisco Bay Area.

Observational data analysis to support decisions in uncertain causal contexts, using structural causal models.

Kiwi Power (*Apr. to Jul. 2016*)

Data Scientist (part-time), London, UK.

Machine learning and econometric methods in R to predict client behaviour, data migration from SQL to MongoDB, interactive dashboards in R Shiny.

The Outside View (*Aug. 2014 to Nov. 2015*)

Data Scientist, London, UK.

Machine learning and statistical methods for real estate, automotive markets and recruitment, with end-to-end analytics solutions in Python.

Product Madness (*Feb. to Aug. 2014*)

Data Scientist, London, UK.

Predictive analytics in Python, imbalance-handling methods, A/B testing for engagement and revenue, dashboards with MySQL, AWS, Tableau and D3.js.

CONFERENCE AND WORKSHOP PRESENTATIONS

CausNetS: Toward a Causal Network Science, NetSci 2026 Satellite (*Jun. 2026*)

Boston, MA. Talk: "Educational Intervention Can Rewire Village Social Networks."

Joint Statistical Meetings (JSM) (*2019, 2022 to 2025*)

JSM 2025, Nashville: "Wavelet Latent Position Exponential Random Graphs." JSM 2024, Portland: "Decision-Theoretic Robustness for Network Models." JSM 2023, Toronto: "State-Space Modeling of Time-Varying Spillovers on Networks." JSM 2022, Washington DC: "Graphon-Level Bayesian Predictive Synthesis for Random Networks." JSM 2019, Denver: "CoLaS: Copula-Seeded Sparse Local Graphs with Tunable Assortativity, Persistent Clustering, and a Degree-Tail Dichotomy."

Sunbelt 2025, International Network for Social Network Analysis (*Jun. to Jul. 2025*)

Paris, France. Two papers, on self-reported error in sociocentric mapping and on social networks and attitudes to domestic violence.

Novel AI and Data for Disaster Management Workshop (*Jul. 2025*)

Computational Social Science Lab, University of Pennsylvania, with Meta AI for Good and Direct Relief's CrisisReady initiative. Philadelphia, PA.

Workshop on the Frontiers in Data Science (*Jan. 2020*)

Keio University, Tokyo, Japan.

International Conference on Scientific and Statistical Database Management (SSDBM) (*Jun. 2012*)

Chania, Crete, Greece. Best Paper Award.

TECHNICAL SKILLS AND TRAINING

Programming: Python, R, Stan, Julia, MATLAB, C/C++, Java, SQL, JavaScript.

Statistical and machine learning: Bayesian modeling, MCMC, variational inference, scikit-learn, PyTorch, TensorFlow.

Large-scale computing: Hadoop, Spark, high-performance computing clusters.

Other: Stata, SAS, SPSS, L^AT_EX, Git, R Shiny.

Training: Cloudera Developer Training for Apache Spark, funded by University College London (2015).

LANGUAGES

English: Fluent. **Greek:** Native. **Spanish:** Professional working proficiency.

REFERENCES

Prof. Nicholas A. Christakis

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